Frontend Migration Guide: API Contract Updates

Transitioning from newcp.new.openapi.yaml (Draft) to newcp.openapi.yaml (Canonical)

Target Audience: Frontend Engineers (New CP React Apps & TS SDK)	**Canonical Spec:** openapi/newcp.openapi.yaml
Migration Impact: **HIGH (Breaking Answer Payloads & Type Strictness)**	**Status:** Approved & Canonical (October 2026)

EXECUTIVE SUMMARY & IMMEDIATE ACTION REQUIRED:
The canonical OpenAPI contract (openapi/newcp.openapi.yaml) replaces the earlier draft (openapi/newcp.new.openapi.yaml). All frontend engineers must adapt client code to two main shifts:
1. **Unified FormAnswer Model (BREAKING):** All legacy carrier fields (optionCode, valueText, valueNumber, valueBoolean, valueDate) have been completely removed. Payloads now strictly accept only fieldCode, optionCodes: string[] (for choice fields), and value (polymorphic string|number|boolean for all other fields). Unknown fields return **400 Bad Request**.
2. **24 Guaranteed ServiceItem Properties:** ServiceItem.required now marks all 24 runtime and DB columns required. TypeScript client generation will no longer produce optional (?) flags for quantities, money amounts, timestamps, or billing capability flags.

1. Summary of Changes at a Glance

Contract Area	Draft (newcp.new.openapi.yaml)	Canonical (newcp.openapi.yaml)	Action Required by Frontend
Choice Fields (single & multi)	Allowed optionCode (string) or optionCodes (array). Multi-select only saved first option.	Strictly optionCodes: string[] for BOTH single and multi-select. optionCode is removed.	**Breaking:** Always send array. Single-select sends ['code']. Multi-select sends ['a', 'b'].
Free-Form Fields (text, number, date, bool)	Separate carrier fields: valueText, valueNumber, valueBoolean, valueDate.	Single polymorphic attribute: value: string | number | boolean.	**Breaking:** Replace all carrier fields with value. Send real JSON primitives (no quoted numbers).
Schema Strictness (additionalProperties)	Not specified (allowed extra/carrier attributes to pass unvalidated).	additionalProperties: false enforced on FormAnswer schema.	**Breaking:** Sending legacy fields (e.g. valueText) now fails with **400 Bad Request**.
ServiceItem Fields	Only 8 fields marked required. Many runtime-guaranteed columns were typed optional (?).	24 fields strictly marked required in ServiceItem.required matching PostgreSQL V3.	**Refinement:** Regenerate TypeScript client. Remove defensive item.quantity ?? 1 fallbacks.
Payer & Billing Flags	Uncertain presence of canEditBilledParty and billedPartyOrigin.	Guaranteed: billedToContactUuid, billedPartyOrigin, canEditBilledParty.	Use canEditBilledParty to toggle Bill-To UI selector. Treat billedPartyOrigin as read-only.

2. Deep Dive: FormAnswer Migration (Payload Changes)

Form answers are submitted when creating (POST /orders/{orderUuid}/services) or updating (PATCH /services/{serviceItemUuid}) service items. The backend has unified the structure into a clean, two-path model based strictly on field type.

Field Type	OLD DRAFT PAYLOAD (newcp.new)	NEW CANONICAL PAYLOAD (newcp.openapi)
Text Field (e.g., special_notes)	`{"fieldCode": "notes", "valueText": "Glaze buccal"}`	`{"fieldCode": "notes", "value": "Glaze buccal"}`
Number Field (e.g., implant_count)	`{"fieldCode": "qty", "valueNumber": 3}`	`{"fieldCode": "qty", "value": 3}` *Note: must be JSON number, not "3"*
Boolean Field (e.g., is_urgent)	`{"fieldCode": "urgent", "valueBoolean": true}`	`{"fieldCode": "urgent", "value": true}` *Note: must be JSON boolean, not "true"*
Date Field (e.g., target_date)	`{"fieldCode": "due", "valueDate": "2026-11-05"}`	`{"fieldCode": "due", "value": "2026-11-05"}` *ISO 8601 YYYY-MM-DD string*
Single Select (e.g., arch_selection)	`{"fieldCode": "arch", "optionCode": "mandibular"}`	`{"fieldCode": "arch", "optionCodes": ["mandibular"]}` *Single element array*
Multi Select (e.g., retention_features)	`{"fieldCode": "features", "optionCodes": ["ball", "loc"]}` *(Draft bug: only saved 1st option)*	`{"fieldCode": "features", "optionCodes": ["ball", "loc"]}` *All options saved and restored properly!*
Clearing Choice (user deselects)	`{"fieldCode": "arch", "optionCodes": []}`	`{"fieldCode": "arch", "optionCodes": []}`

CRITICAL VALIDATION RULES FOR FRONTEND SERIALIZATION:

- **Exactly one payload path:** Send optionCodes for choice fields; send value for free-form fields. Never send both.
- **No legacy carrier fields:** Do NOT send optionCode, valueText, valueNumber, etc. additionalProperties: false triggers **400 Bad Request**.
- **Type mismatch enforcement:** Sending value for a choice field or optionCodes for a text/number field triggers **422 INVALID_FIELD_TYPE**.
- **Full replacement on PATCH:** The answers array in PATCH /services/{uuid} completely replaces all existing answers. Always send the full active set.

3. Reading Answers from Server Responses

When fetching service items (GET /services/{uuid}) or receiving responses from create/patch endpoints, the returned answers[] array uses the identical simplified format. Update client-side form population logic as follows:

```
// TypeScript helper to populate form state from API response
interface FormAnswerDto {
  fieldCode: string;
  value?: string | number | boolean;
  optionCodes?: string[];
}

function populateFormState(fieldDefinitions: FormField[], answers: FormAnswerDto[]) {
  const answerMap = new Map(answers.map(a => [a.fieldCode, a]));

  return fieldDefinitions.map(field => {
    const answer = answerMap.get(field.code);
    if (field.type === 'single_select') {
      // Extract single option code from optionCodes array
      return { code: field.code, selected: answer?.optionCodes?.[0] ?? null };
    } else if (field.type === 'multi_select') {
      // Extract all option codes
      return { code: field.code, selected: answer?.optionCodes ?? [] };
    } else {
      // Free-form fields: bind directly to polymorphic value
      return { code: field.code, value: answer?.value ?? field.defaultValue ?? '' };
    }
  });
}
```

4. ServiceItem Type Guarantees (24 Required Fields)

In the draft contract, only 8 fields were in ServiceItem.required. Frontend developers had to write defensive fallback checks (e.g. item.quantity ?? 1 or item.isBillable ?? true). In newcp.openapi.yaml, the required list has been expanded to **24 fields** matching the PostgreSQL V3 database constraints:

Category	Required Field Name	Type	Frontend Purpose & Usage
Identity & Order	`uuid, orderUuid, sellableItemUuid, lineNumber`	UUIDs & integer	Order & catalog linkage. lineNumber is now guaranteed integer (item display order on order screen).
Lifecycle & Status	`status, isRedo, isInvoiced`	string & booleans	Item lifecycle stage. isInvoiced: true locks financial edits across all UI controls.
Financials & Totals	`currencyCode, unitPrice, fieldsAmount, discountAmount, voucherAmount, lineTotal, quantity`	strings & numbers (2/3 dec)	Complete money breakdown. quantity is fractional (>= 0.001). fieldsAmount reflects custom field surcharges.
Billing & Payer Controls	`isBillable, billedToContactUuid, billedPartyOrigin`	boolean, UUID, string enum	**Key UI controls:** isBillable is independent of payer. billedPartyOrigin is read-only server provenance.
UI Capability Flags	`canEditBilledParty, canEditPrice, canEditBillability`	booleans	**Directly bind to UI controls!** If canEditBilledParty is false, disable the payer dropdown.
Metadata & Concurrency	`answers, etag, createdAt, updatedAt`	array, string, ISO dates	etag is required for optimistic locking headers. answers is always an array (never undefined).

5. Billed Party Selection Flow in UI

The New CP backend enforces tenant security and per-item billing. Frontend applications should implement the payer selection UI using the dedicated endpoint:

Payer Selector Implementation Steps:
1. **Fetch Options:** Call GET /orders/{orderUuid}/eligible-billed-parties to populate the payer dropdown. Returns doctor and scanning center contacts associated with the order (never an unrestricted contact list).
2. **Preselection:** When creating a service item (POST /orders/{orderUuid}/services), omit billedToContactUuid to automatically default to the order-level default contact, or pass an explicit contact UUID.
3. **Editing Payer:** On existing items, inspect canEditBilledParty. If false, display the party as read-only. If true, allow the user to select an eligible contact and send PATCH /services/{uuid} with billedToContactUuid.
4. **Never Send Provenance:** Do not include billedPartyOrigin in request bodies. The server assigns and audits provenance.

6. Concurrency Locking & Workflow Transitions

Operation	Endpoint & Method	Required Headers	Failure Status Codes
Update Service Details (answers, payer, price)	`PATCH /services/{serviceItemUuid}`	If-Match: "<etag>" (Use current item.etag)	**428 Precondition Required** (missing header) **412 Precondition Failed** (stale item)
Execute Transition (status change, hold)	`POST /services/{serviceItemUuid} /transitions`	If-Match: "<etag>" (Use current item.etag)	**428 Precondition Required** (missing header) **412 Precondition Failed** (stale item)

7. Frontend Developer Migration Checklist

Step	Action Item	Files / Components Typically Affected	Status
1	**Regenerate SDK:** Run openapi-generator / openapi-typescript using openapi/newcp.openapi.yaml.	src/api/client.ts, schema.d.ts	[ ] Pending
2	**Update Form Answer Serializer:** Map choice fields to optionCodes: string[] and all others to value.	FormRenderer.tsx, answerHelpers.ts	[ ] Pending
3	**Purge Carrier Fields:** Verify no code sends optionCode, valueText, valueNumber, etc.	Form submission hooks, Redux / React Query mutations	[ ] Pending
4	**Update Form Answer Reader:** Read answer.optionCodes for choice fields and answer.value for free-form.	Service item detail view, Edit Form dialog	[ ] Pending
5	**Remove Defensive Fallbacks:** Take advantage of 24 required fields on ServiceItem (clean up ?? operators).	Service table columns, Pricing summaries	[ ] Pending
6	**Bind UI Permission Flags:** Disable payer selector when canEditBilledParty === false; disable price input when canEditPrice === false.	BilledPartySelector.tsx, PriceInput.tsx	[ ] Pending
7	**Verify If-Match Header:** Ensure PATCH and transitions attach If-Match: "<etag>" and handle 412 refresh prompt.	API interceptor, Mutation error handlers	[ ] Pending